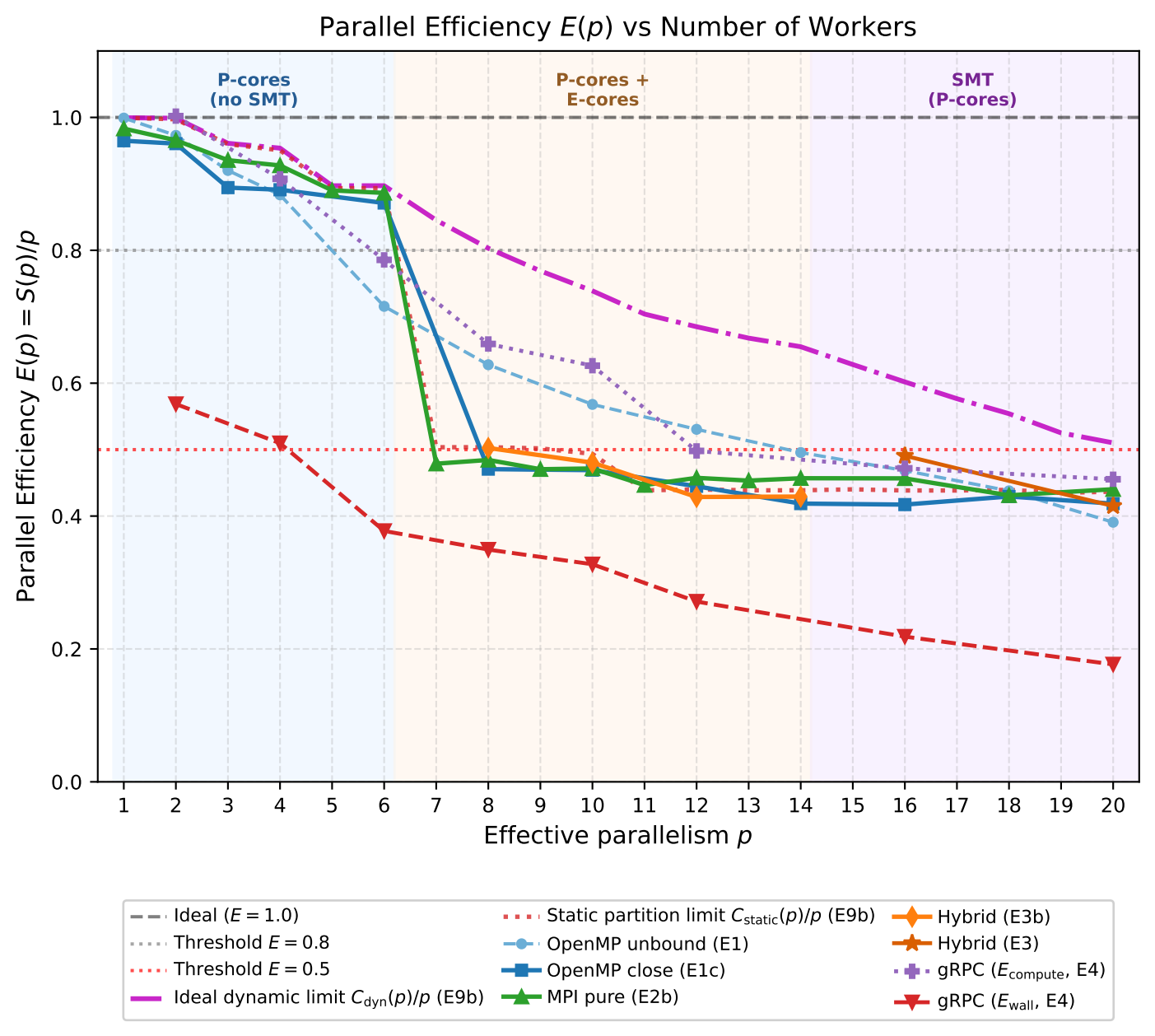
Parallel Efficiency $E(p)$ vs Number of Workers
P-cores (no SMT)
P-cores + E-cores
SMT (P-cores)
Parallel Efficiency $E(p) = S(p)/p$
Effective parallelism p
1.0
0.8
0.6
0.4
0.2
0.0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
Ideal ($E = 1.0$)
Threshold $E = 0.8$
Threshold $E = 0.5$
Ideal dynamic limit $C_{dyn}(p)/p$ (E9b)
Static partition limit $C_{static}(p)/p$ (E9b)
OpenMP unbound (E1)
OpenMP close (E1c)
MPI pure (E2b)
Hybrid (E3b)
Hybrid (E3)
gRPC ($E_{compute}$, E4)
gRPC (E_{wall}, E4)